Responses to Reviewers' Comments for Manuscript COMNET-D-26-06580

Joint Horizontal-Vertical Speed Optimization for Energy-Efficient UAV Data Collection

Addressed Comments for Publication to

Computer Networks

by

Haozhe Zhang, Feng Shan and Ming Yang

Dear Editor,

Please find enclosed the revised version of our previous submission entitled "Joint Horizontal-Vertical Speed Optimization for Energy-Efficient UAV Data Collection" with manuscript number COMNET-D-26-06580. We would like to thank you and the reviewers for the valuable comments, which have helped improve the quality of our manuscript. In this revision, we have carefully addressed the reviewers' comments. A summary of the main modifications and a detailed point-by-point response to the comments from Reviewers 1 and 2 (following the reviewers' order in the decision letter) are given below.

Sincerely,

Haozhe Zhang, Feng Shan and Ming Yang

Note: To enhance the legibility of this response letter, all the editor's and reviewers' comments are typeset in boxes. Rephrased or added sentences are typeset in color. The respective parts in the manuscript are highlighted to indicate changes.

Response to the Meta Review

Summary Comment 1

The editor has collected the review comments from two reviewers, and both reviewers recommend major revision. The manuscript should strengthen the data-collection and propulsion models and better clarify the claimed coupling between horizontal and vertical motion. The practical feasibility of the trajectory should also be improved by addressing velocity transitions and the possible impact of wind. In addition, the global optimality claim should be qualified since the proposed DP operates over a discretized state space. Further analysis of discretization sensitivity and computational cost is recommended. The authors should also discuss the assumptions related to mission duration and practical UAV operating constraints. Based on these concerns, the editor recommends major revision for this submission.

We sincerely thank the Editor for the careful assessment and for synthesizing the reviewers' concerns. We have revised the manuscript to address every point raised above. A brief summary of how each concern has been resolved is given below; the detailed arguments and the corresponding manuscript changes are provided in our point-by-point responses to the reviewers.

- **Strengthening the data-collection and propulsion models.** We have enriched the data-collection model by expressing the data upload through a constant achievable rate determined by the channel bandwidth and a per-node nominal SINR, and by tying the required collection duration to the data volume (Comment 1.1); the angular unit of the LoS model has also been clarified (Comment 2.1). The propulsion model now states explicitly how the horizontal and vertical velocity components combine into the total speed (Comment 1.2).
- **Clarifying the claimed coupling between horizontal and vertical motion.** We have revised the wording so that the jointness is stated precisely—through the flight pitch angle, which ties the two velocity components together—rather than over-claiming a coupling in the instantaneous power itself, which is additive under the adopted models. Please see our response to Comment 1.2.
- **Practical feasibility of the trajectory (velocity transitions and wind).** We have added a discussion of the acceleration/deceleration cost at segment boundaries (Comments 1.3 and 2.2) and of the effect of wind, *i.e.,* the distinction between airspeed and ground speed (Comment 1.4).
- **Qualifying the global-optimality claim.** We have qualified the optimality statements as holding *over the discretized state space* (e.g., in the conclusion), and demonstrated that the discretized optimum converges to the continuous optimum with negligible loss. Please see our response to Comment 1.5.
- **Discretization sensitivity and computational cost.** We have added a new sensitivity study that varies the state-sampling resolution and reports the energy convergence, the running time, and the relative optimality gap. Please see our response to Comment 1.5.
- **Assumptions on mission duration and operating constraints.** We have added explicit discussions of the no-global-deadline assumption (Assumption A2), the altitude bounds, and the battery-capacity constraint. Please see our responses to Comments 2.3, 2.4 and 2.5.

Response to Reviewer 1

Comment 1.1

The data-collection model is overly simplified. Although each GN has a certain data volume, successful collection is determined only by the path-loss threshold and the prescribed duration τ_i, without explicitly modeling achievable rate, bandwidth, SINR, or data size.

Response 1.1:

Thank you for this valuable comment. We agree that the previous formulation left the relationship between the prescribed collection duration τ_i and the underlying data-collection process implicit. We have revised the system model to explicitly incorporate the achievable rate, the bandwidth, the SINR, and the data volume, and to tie the successful-collection condition to these quantities.

(1) Nominal SINR and achievable rate. In the revised Section 3.1, we characterize each GN_i by a nominal SINR γ_i, a per-node design parameter given in advance (like the path-loss threshold PL_i^{max}), rather than the instantaneous, position-dependent SINR experienced by the UAV. The achievable rate is evaluated at this nominal SINR through the Shannon capacity, $\bar{R}_i = B\log_2(1+\gamma_i)$, where B is the channel bandwidth.

(2) Constant-rate characterization. Since γ_i is a fixed per-node parameter, the achievable rate $\bar{R}_i$ is constant throughout the collection phase, which keeps the collection duration $\tau_i = D_i/\bar{R}_i$ a fixed per-node quantity.

(3) Successful-collection condition. Given the data volume D_i of GN_i, the required collection duration is now explicitly $\tau_i = D_i/\bar{R}_i$, and the collection of GN_i is defined to be successful if and only if the UAV stays within its communication region for a total duration of at least τ_i. This connects the path-loss threshold (which defines the coverage region) to the data volume, bandwidth, and SINR through the achievable rate, while preserving the fixed-τ_i structure of the problem.

The simulation setup is updated accordingly: $B = 1$ MHz, γ_i sampled per GN from $[12, 18]$ (default 15), and D_i averaged 88 Mbit with range $[72, 100]$ Mbit, so that $\tau_i = D_i/\bar{R}_i \approx 22$ s on average (matching the original configuration).

The revised text in the manuscript is as follows.

Section 3.1 (added). To model the data upload, we characterize each GN_i by a nominal SINR γ_i, which is a per-node design parameter given in advance—like the path-loss threshold PL_i^{max}—rather than the instantaneous, position-dependent SINR experienced by the UAV. Letting B denote the channel bandwidth, the achievable rate of the link is evaluated at this nominal SINR through the Shannon capacity:

$$\bar{R}_i = B\log_2(1+\gamma_i).$$

Each GN_i therefore uploads data at the constant rate $\bar{R}_i$ throughout the collection phase, which yields the deterministic collection duration $\tau_i = D_i/\bar{R}_i$.

Section 3.3 (revised Definition). Given ordered ground nodes $GN_1, \ldots, GN_N$, their communication regions, and the data volume D_i of each GN_i, the *JHVSO* problem seeks a trajectory $q(t) = (d(t), h(t))$ and collection intervals that minimize the total

flight energy. Under the constant achievable rate $\bar{R}_i$, collecting the data volume D_i requires a duration $\tau_i = D_i/\bar{R}_i$; the collection of GN_i is successful if and only if the UAV stays within its communication region for a total duration of at least τ_i.
Table 1 (revised). The row "Average τ (s) 22 [18 25]" is replaced by "Average D_i (Mbit) 88 [72 100]", and a new row "Nominal SINR γ_i 15 [12 18]" is added.
Section 6 (added). The data upload adopts a channel bandwidth of $B = 1$ MHz and a per-GN nominal SINR γ_i sampled from $[12, 18]$ (default 15), which yields a constant achievable rate $\bar{R}_i = B\log_2(1+\gamma_i)$, about 4 Mbps at the default; the required collection duration is then $\tau_i = D_i/\bar{R}_i$, approximately 22 s on average under the default data volume $D_i = 88$ Mbit.
Figure 2 (revised caption). The UAV must remain within the region of GN_i for the required data collection duration $\tau_i = D_i/\bar{R}_i$, where D_i is the data volume of GN_i and $\bar{R}_i = B\log_2(1+\gamma_i)$ is the constant achievable rate.

Comment 1.2

The claimed coupling between horizontal and vertical propulsion is not sufficiently reflected in the model as shown in Eq. (3). The total power is formulated as the additive form. The authors should clarify what physical coupling is actually captured.

Response 1.2:

Thank you for this comment. We agree that the horizontal and vertical power models are additive and therefore do not exhibit a coupling between the two power components; the earlier phrase "continuous coupling between horizontal and vertical propulsion power" was an imprecise wording, which we have now corrected throughout the manuscript.

To state the relationship between the two velocity components precisely, we have added, immediately before the total power model in Eq. (4), the relation

$$V_{\parallel} = V_{total}\cos\gamma, \qquad V_{\perp} = V_{total}\sin\gamma,$$

where V_{total} is the total flight speed and γ the flight pitch angle. This relation expresses how the horizontal and vertical components combine into the total speed and, through the additive power model, into the total power. Accordingly, the wording "coupling between horizontal and vertical propulsion power" has been replaced by "joint horizontal–vertical speed optimization".

The revised text is as follows.

Abstract (revised). To capture realistic three-dimensional motion, we adopt a practical 3D flight energy consumption model and jointly optimize the horizontal and vertical speeds.
Introduction (revised). Instead of optimizing horizontal speed and height separately, we focus on joint horizontal-vertical speed optimization.
Related work (revised). However, existing 3D extensions typically schedule the two dimensions independently (e.g., as discrete altitude transitions). In contrast, our formulation retains the additive instantaneous-power form of [11,12] but jointly optimizes the two speed components through the flight pitch angle, as detailed in Section 3.2.

Section 3.2 (added). Let V_{total} denote the total (resultant) flight speed and γ the flight pitch angle, defined as the angle between the velocity vector and the horizontal plane. The horizontal and vertical velocity components are related to V_{total} and γ by

$$V_{\parallel} = V_{total}\cos\gamma, \qquad V_{\perp} = V_{total}\sin\gamma.$$

Comment 1.3

Regarding the Theorem in Section 3.2, the resulting trajectory may not be dynamically feasible for a physical UAV. Consecutive flight segments can have different velocities at their shared boundary, effectively requiring instantaneous velocity changes. The authors should discuss whether such piecewise constant-speed trajectories can be practically tracked and consider incorporating acceleration constraints or transition dynamics into the model.

Response 1.3:

Thank you for this comment. The optimal trajectory is indeed piecewise constant-speed, so consecutive segments may differ in velocity at their shared boundary. We have clarified this point in the revised manuscript as follows.

The constant segment speeds should be interpreted as target cruise speeds, which a physical UAV tracks with short acceleration/deceleration ramps. The time and energy consumed during these transition phases account for only a small fraction of the total mission time and energy [1], so they do not materially affect the planned schedule or the energy-optimal speed selection. We have added a remark in Section 4.1 stating this explicitly; incorporating full acceleration dynamics into the optimality analysis is left as future work, consistent with the existing limitation in the conclusion.

The revised text is as follows.

Section 4.1 (added Remark). The resulting trajectory is piecewise constant-speed, with task segments flown per Remark 1 and gap segments per Theorem 2, so consecutive segments may change velocity at their shared boundary. In practice, this change is realized by a brief acceleration/deceleration phase whose time and energy are only a small fraction of the total mission [1]; the constant speeds are therefore target cruise speeds tracked with short transition ramps, and full acceleration dynamics are left as future work.

Comment 1.4

Wind and other environmental disturbances are ignored. Since propulsion power depends on airspeed while trajectory progression depends on ground speed, wind may significantly affect both the energy consumption and the optimal speed schedule.

Response 1.4:

Thank you for this insightful comment. We agree that wind changes the airspeed relative to the ground speed, and thus affects both the propulsion power and the energy-optimal speed

schedule. As the present work focuses on establishing the foundational joint horizontal–vertical speed optimization framework under nominal conditions, we have deferred the explicit modeling of wind to future work, and we have stated this point in the revised conclusion.

The revised text is as follows.

> **Section 7 (revised).** Second, the communication model uses the expected path loss as a deterministic threshold, and environmental disturbances such as wind are not considered. A steady wind, for instance, shifts the UAV's airspeed relative to its ground speed, thereby altering both the propulsion power and the energy-optimal speed schedule; explicitly modeling wind and stochastic channel conditions is an important direction for future work.

Comment 1.5

The claimed global optimality is only guaranteed over the discretized state space. The manuscript should provide a sensitivity study on the discretization resolution, energy convergence, and computational cost.

Response 1.5:

Thank you for this important comment. We have addressed it in two respects: we have qualified the global-optimality claim throughout the manuscript, and we have added a dedicated sensitivity study.

(1) Qualifying the optimality claim. All optimality statements are now explicitly scoped to the discretized state space. In particular, the abstract and Theorem 4 state that the ORE algorithm guarantees global optimality *over the discretized state space*, and the conclusion no longer claims optimality over the continuous trajectory design.

(2) Sensitivity study. We have added a new Section 6.3, *Discretization Sensitivity and Computational Cost*, which varies the state sampling cap $M \in \{250, 500, 1000, 2000, 4000\}$ and reports (i) the energy convergence, (ii) the running time, and (iii) the relative optimality gap with respect to the finest resolution $M = 4000$ (new Fig. 9).

(3) Findings. The optimal energy decreases monotonically and converges as M grows: refining M from 1000 to 4000 reduces the energy by only 0.14%, and the improvement roughly halves whenever the resolution doubles, indicating first-order convergence. At the operating resolution $M = 2000$, the optimal energy is within 0.07% of that at $M = 4000$, at an average running time of 4.86 s. Hence a sampling cap in the range $M \in [1000, 2000]$ provides a good balance between optimality and computational cost, confirming that the discretized optimum has essentially converged to the continuous optimum.

The added text in the revised manuscript is as follows.

> **Section 6.3 (added).** The global optimality guarantee of the proposed ORE algorithm holds over the discretized state space, i.e., over the finite set of candidate states sampled within the communication region of each GN. Concretely, the continuous communication region of each GN is first rasterized onto a fine grid in the (d, h) plane with a resolution of 1 m, which yields the full set of candidate state points located inside

the coverage region. When a GN contains more candidate points than a prescribed cap M, the points are uniformly subsampled in raster order—keeping every s-th point with $s = \lfloor N_c/M \rfloor$, where N_c is the number of candidate points of that GN—so that approximately M states remain; GNs whose coverage contains no more than M candidate points keep all of them. As a result, the actual number of sampled states of each GN fluctuates around M depending on its coverage area (on average approximately 2357 and 5791 states for $M = 2000$ and $M = 4000$, respectively). To assess the impact of this sampling resolution, we evaluate the algorithm under different caps $M \in \{250, 500, 1000, 2000, 4000\}$.

As shown in Fig. 9(a), the optimal energy decreases monotonically and converges as M increases: refining M from 1000 to 4000 reduces the total energy by only 0.14%, and the improvement shrinks as the resolution doubles, indicating approximately first-order convergence. At the operating resolution $M = 2000$, the optimal energy is within 0.07% of that obtained at $M = 4000$ (Fig. 9(c)); this residual gap reflects the discretization error of the state space and is negligible in practice, confirming that the discretized optimum has essentially converged to the continuous optimum. Meanwhile, as shown in Fig. 9(b), the running time grows with M, averaging 4.86 s at the operating resolution $M = 2000$. Overall, a sampling cap in the range $M \in [1000, 2000]$ strikes a good balance between energy optimality and computational cost: the optimality gap is already only 0.14% at $M = 1000$ and 0.07% at $M = 2000$, while the running time remains modest. This confirms that the global-optimality guarantee over the discretized state space carries over to the continuous problem with negligible loss.

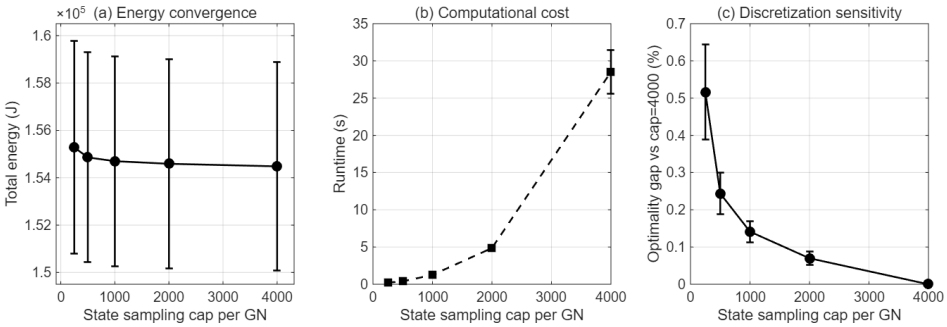


Figure 9 (added). Discretization sensitivity and computational cost: (a) total energy, (b) running time, and (c) the relative optimality gap versus the state sampling cap M.

Section 7 (added sentence). A sensitivity study further confirms that the discretized optimum converges to the continuous optimum with negligible loss (within 0.07% at the operating resolution).

[1] R. Jia, Q. Fu, Z. Zheng, G. Zhang, and M. Li, "Energy and time trade-off optimization for multi-UAV enabled data collection of IoT devices," *IEEE/ACM Transactions on Networking*, vol. 32, no. 6, pp. 5172–5187, 2024.

Response to Reviewer 2

Comment 2.1

The angular unit used in the LoS probability model should be clarified in Eqs. (1)-(2). The definition $\theta_i = \arctan(h_i/r_i)$ appears potentially inconsistent with the simulation parameter $a = 45$ unless the angle is explicitly converted to degrees.

Response 2.1:

Thank you for this comment. The elevation angle in the LoS probability model is expressed in degrees, which is consistent with the standard probabilistic LoS model [1] and with the simulation setting $a = 45$. To remove the ambiguity, we have revised the manuscript to write the angle explicitly in degrees, i.e., $\theta_i = \frac{180}{\pi}\arctan(h_i/r_i)$, and to unify the notation of the environment parameters as a and b throughout Section 3.1.

The revised text is as follows.

Eq. (1) (revised). $PL_i(r_i, h_i) = \frac{\mu_{LoS}-\mu_{NLoS}}{1+a\exp(-b[\frac{180}{\pi}\arctan(h_i/r_i)-a])} + 10\log(h_i^2 + r_i^2) + M.$

Section 3.1 (revised). The key angular variable is the link elevation angle $\theta_i = \frac{180}{\pi}\arctan(h_i/r_i)$ (in degrees), measured between the GN–UAV line and the horizontal plane. It determines the LoS probability $P_{LoS}(\theta_i) = 1/(1 + a\exp(-b[\theta_i - a]))$.

Comment 2.2

The propulsion model ignores acceleration and deceleration costs in Theorem 1. The UAV can therefore switch between different segment velocities without any transition time or energy penalty, which may affect the practical validity of the obtained trajectory.

Response 2.2:

Thank you for this comment. We have clarified this point in the revised manuscript: the constant segment speeds should be interpreted as target cruise speeds, which a physical UAV tracks with short acceleration/deceleration ramps. Moreover, the time and energy consumed during these transition phases account for only a small fraction of the total mission time and energy [2], so they do not materially affect the planned schedule or the energy-optimal speed selection. We have added a remark in Section 4.1 stating this explicitly, and full acceleration dynamics is left as future work (also noted in the conclusion).

The revised text is as follows.

Section 4.1 (added Remark). The resulting trajectory is piecewise constant-speed, with task segments flown per Remark 1 and gap segments per Theorem 2, so consecutive segments may change velocity at their shared boundary. In practice, this change is realized by a brief acceleration/deceleration phase whose time and energy are only a small fraction of the total mission [2]; the constant speeds are therefore target cruise speeds tracked with short transition ramps, and full acceleration dynamics are left as future work.

Comment 2.3

In Theorem 3, the non-aftereffect property relies on the strong assumption that there is no global mission deadline. With a practical constraint such as $T \le T_{\max}$, elapsed time would need to be included in the DP state.

Response 2.3:

Thank you for this comment. We respond in two parts.

(1) The non-aftereffect property and the essential role of Assumption A2. The mission is decomposed into per-GN stages, and the cost of each inter-node transition is determined solely by its two endpoint states; Theorem 3 shows that the optimal outbound sub-path from any state S depends only on S, not on the inbound path. This non-aftereffect property rests on Assumption A2 (no global mission deadline): without a deadline, the elapsed time does not constrain the feasibility of the outbound path, so the state S alone is a sufficient statistic for all future decisions. The assumption is deliberate—with a global deadline $T \le T_{\max}$, the remaining time budget would depend on the elapsed time, which would then have to enter the DP state and break the low-dimensional Markov structure of Theorem 3. We have added a remark in Section 4.2 stating this explicitly.

(2) Future work with a deadline. Should a global deadline be imposed, each DP state would need to be augmented with a time dimension, and the overall algorithm would have to be revisited accordingly; we leave this as an important direction for future work.

The revised text is as follows.

Section 4.2 (added Remark). Assumption A2 is essential to Theorem 3: with a global deadline $T \le T_{\max}$, the feasibility of an outbound path would depend on the elapsed time, so each DP state would need to be augmented with a time dimension, and the overall algorithm would have to be revisited accordingly. The no-deadline assumption is therefore deliberate, as it keeps the state space low-dimensional and enables the efficient stage-wise recurrence; extending the framework to deadline-constrained missions is left as future work.

Comment 2.4

Practical altitude constraints are not explicitly considered. Constraints such as $h_{\min} \le h(t) \le h_{\max}$, terrain clearance, obstacles, and regulatory altitude limits should be discussed.

Response 2.4:

Thank you for this comment. We have added explicit altitude bounds to the formulation as follows.

First, the flight altitude is now bounded by $h_{\min} \le h(t) \le h_{\max}$ in Definition 1. Second, this bound is enforced in a straightforward manner: candidate state points whose altitude violates the bound are simply discarded during state-space construction, so the dynamic program only ever considers altitude-feasible states, and no other change to the algorithm is required. Third, in the simulations the altitude is confined to $[50, 150]$ m—the lower bound matches the take-off/landing altitude, while the upper bound is a representative operating ceiling for infrastructure-inspection missions, which permits altitudes above the 120-m open-category

limit when the UAV stays in close proximity to a structure (14 CFR §107.51); it also covers the highest altitude (135 m) reached in our real-world case study. Terrain-clearance and obstacle constraints, which amount to position-dependent forbidden regions, are not modeled and are left as a direction for future work.

The revised text is as follows.

> **Definition 1 (revised).** The flight altitude is additionally bounded by $h_{\min} \le h(t) \le h_{\max}$, where $h_{\min}$ and $h_{\max}$ are the lower and upper altitude limits.
> **Section 4.2 (added).** Candidate state points whose altitude violates $h_{\min} \le h \le h_{\max}$ are simply discarded during state-space construction, so the dynamic program only ever considers altitude-feasible states; no other change to the algorithm is required.
> **Section 6 (added).** The flight altitude is confined to $[h_{\min}, h_{\max}] = [50, 150]$ m: the lower bound matches the take-off/landing altitude, while the upper bound is a representative operating ceiling for infrastructure-inspection missions (e.g., 14 CFR §107.51), and it covers the highest altitude (135 m) in our real-world case study.
> **Table 1 (added row).** Altitude bounds $[h_{\min}, h_{\max}]$ (m) [50, 150].

Comment 2.5

The formulation does not explicitly consider the UAV battery capacity, e.g., $E_{total} \le E_{battery}$. Moreover, takeoff, initial climb, final descent, and landing energy are excluded from the simulated mission.

Response 2.5:

Thank you for this comment. We respond to the two points separately.

(1) Battery capacity. A battery-capacity constraint $E_{total} \le E_{battery}$ is analogous to the global mission deadline discussed in our response to Comment 2.3: if enforced, the accumulated energy would have to be tracked in the DP state, which would break the low-dimensional Markov structure of Theorem 3. We therefore do not impose it in the present formulation and leave it as a direction for future work (noted in the conclusion).

(2) Takeoff, climb, descent, and landing energy. These energies are fixed one-time costs: since the initial and final altitudes h_0 and h_f are exogenous, they are additive constants that do not affect the optimal trajectory. We have stated explicitly in Section 3.3 that they are excluded from E_{total}.

The revised text is as follows.

> **Section 3.3 (added).** The energies consumed during takeoff, the initial climb to h_0, the final descent from h_f, and landing are fixed one-time costs: since h_0 and h_f are exogenous, these costs are additive constants that do not affect the optimal trajectory, and are therefore excluded from E_{total}.
> **Section 7 (revised).** Finally, the extension to unknown visiting orders, multiple UAVs, online scheduling under uncertain data volumes, explicit terrain/obstacle constraints, and an onboard battery-capacity constraint are promising directions for future work.

[1] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Communications Letters*, vol. 3, no. 6, pp. 569–572, 2014.

[2] R. Jia, Q. Fu, Z. Zheng, G. Zhang, and M. Li, "Energy and time trade-off optimization for multi-UAV enabled data collection of IoT devices," *IEEE/ACM Transactions on Networking*, vol. 32, no. 6, pp. 5172–5187, 2024.